COMP3670/6670
Introduction to Machine Learning
Week 8b: Hyperplanes & Support Vector Machines

Yun Kuen Cheung, Marco
Senior Lecturer, School of Computing, ANU

Goals of this lecture

- Use an affine score to classify a datapoint.
- Turn scores into distances to a hyperplane.
- Understand why an SVM seeks a large margin.
- Allow violations using slack variables and hinge loss.
- See how numerical methods compute the model parameters.

Affine scores classify datapoints

- An email filter predicts spam or not from features such as word frequencies.
- The training data are $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$, with labels $y_i \in \{-1, +1\}$.
- We want positive scores for one class and negative scores for the other.

Affine score. A weight vector $\mathbf{w} \in \mathbb{R}^d$ and a bias $b \in \mathbb{R}$ give the score

$$f_{\mathbf{w},b}(\mathbf{x}) \;=\; \langle \mathbf{w}, \mathbf{x} \rangle + b.$$

The classifier predicts

$$\widehat{y}(\mathbf{x}) \;=\; \begin{cases} +1, & f_{\mathbf{w},b}(\mathbf{x}) \geq 0, \\ -1, & f_{\mathbf{w},b}(\mathbf{x}) < 0. \end{cases}$$

The bias allows the boundary to move away from the origin.

Decision boundary. When $\mathbf{w} \neq \mathbf{0}$, the affine hyperplane

$$\mathcal{H} \;=\; \{\mathbf{x} \in \mathbb{R}^d : \langle \mathbf{w}, \mathbf{x} \rangle + b = 0\}$$

is the classifier's decision boundary. Its normal vector is $\mathbf{w}$.

If $\mathbf{u}, \mathbf{v} \in \mathcal{H}$, then

$$\langle \mathbf{w}, \mathbf{u} - \mathbf{v} \rangle \;=\; -b - (-b) \;=\; 0.$$

Thus $\mathbf{w}$ is perpendicular to every displacement along the boundary.

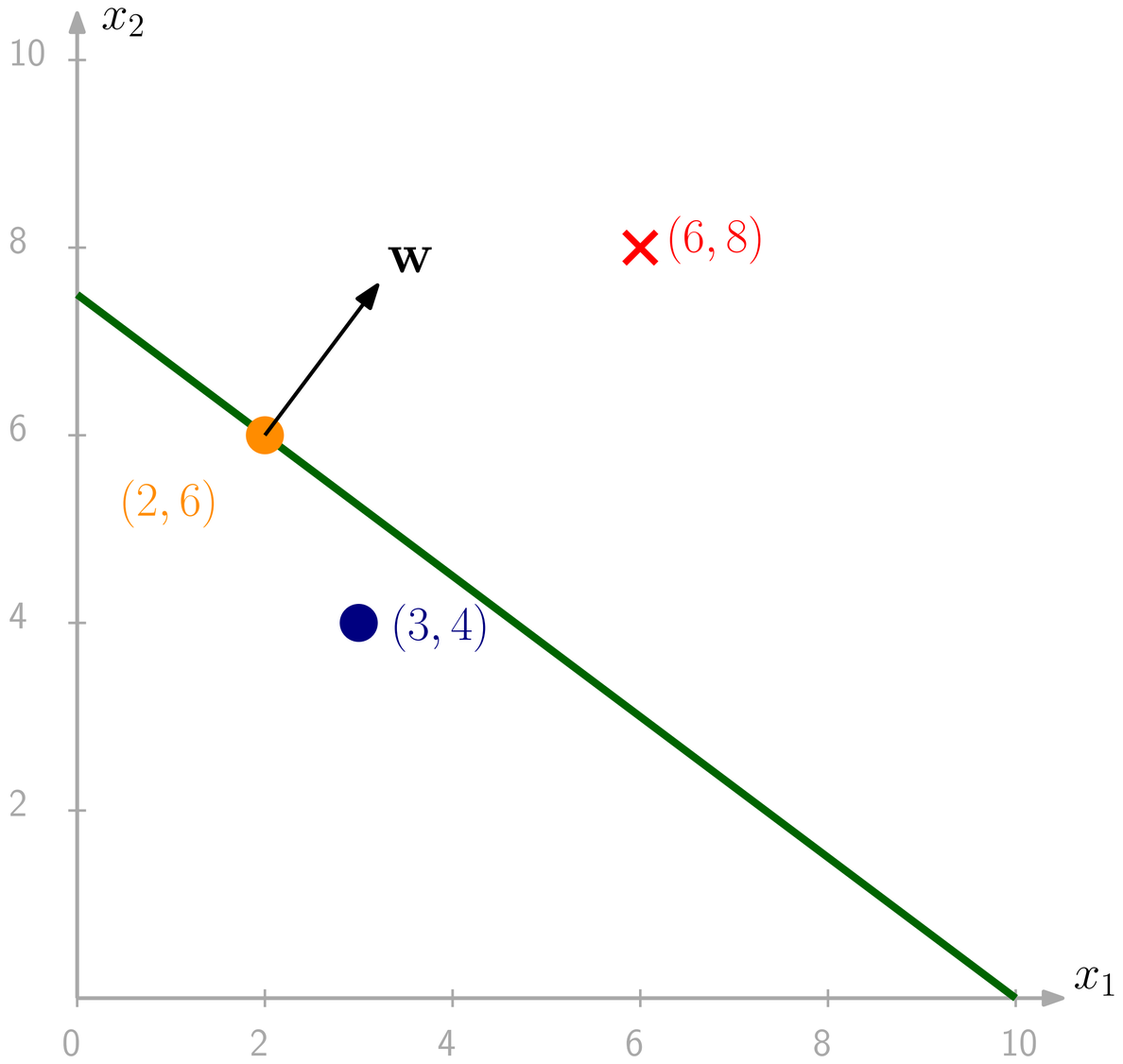
Example 1. Take $\mathbf{w} = (3, 4)^{\mathsf{T}}$ and $b = -30$. Then

$$f_{\mathbf{w},b}(\mathbf{x}) \;=\; 3x_1 + 4x_2 - 30.$$

$\mathbf{x}$	$f_{\mathbf{w},b}(\mathbf{x})$	location
$(6, 8)^{\mathsf{T}}$	20	positive side
$(3, 4)^{\mathsf{T}}$	-5	negative side
$(2, 6)^{\mathsf{T}}$	0	on the boundary

At score zero, our prediction convention gives $+1$.

Example 1 (cont'd). A hyperplane in $\mathbb{R}^2$ is a line.



The boundary is

$$3x_1 + 4x_2 \;=\; 30.$$

Positive side: prediction $+1$.

Negative side: prediction -1.

$\mathbf{w} = (3, 4)^\mathsf{T}$ points perpendicular to the line.

If $b = 0$, the boundary passes through the origin.

Distance gives scores a geometric meaning

- Several hyperplanes may classify all training datapoints correctly.
- A point close to the boundary can change its prediction after a small feature perturbation.
- Multiplying $\mathbf{w}$ and b by a positive constant rescales scores but keeps the boundary fixed.

Proposition 1. For $\mathcal{H} = \{\mathbf{z} : \langle \mathbf{w}, \mathbf{z} \rangle + b = 0\}$ with $\mathbf{w} \neq \mathbf{0}$, the closest point to $\mathbf{x}$ is

$$\mathbf{p} \;=\; \mathbf{x} - \frac{\langle \mathbf{w}, \mathbf{x} \rangle + b}{\|\mathbf{w}\|_2^2}\mathbf{w}.$$

The distance and signed distance are

$$d(\mathbf{x}, \mathcal{H}) \;=\; \frac{|f_{\mathbf{w},b}(\mathbf{x})|}{\|\mathbf{w}\|_2}, \qquad s(\mathbf{x}, \mathcal{H}) \;=\; \frac{f_{\mathbf{w},b}(\mathbf{x})}{\|\mathbf{w}\|_2}.$$

Signed distance is positive on the side towards which $\mathbf{w}$ points.

Proof outline. Verify that $\mathbf{p} \in \mathcal{H}$ and that $\mathbf{x} - \mathbf{p}$ is parallel to $\mathbf{w}$.

For any $\mathbf{z} \in \mathcal{H}$, we have $(\mathbf{x} - \mathbf{p}) \perp (\mathbf{p} - \mathbf{z})$. Hence

$$\|\mathbf{x} - \mathbf{z}\|_2^2 \;=\; \|\mathbf{x} - \mathbf{p}\|_2^2 + \|\mathbf{p} - \mathbf{z}\|_2^2 \;\geq\; \|\mathbf{x} - \mathbf{p}\|_2^2.$$

Equality holds exactly when $\mathbf{z} = \mathbf{p}$.

Finally, take the norm of $\mathbf{x} - \mathbf{p}$ to obtain the distance formula.

Example 2. Use $\mathbf{w} = (3, 4)^{\mathsf{T}}$, $b = -30$ and $\mathbf{x} = (6, 8)^{\mathsf{T}}$.

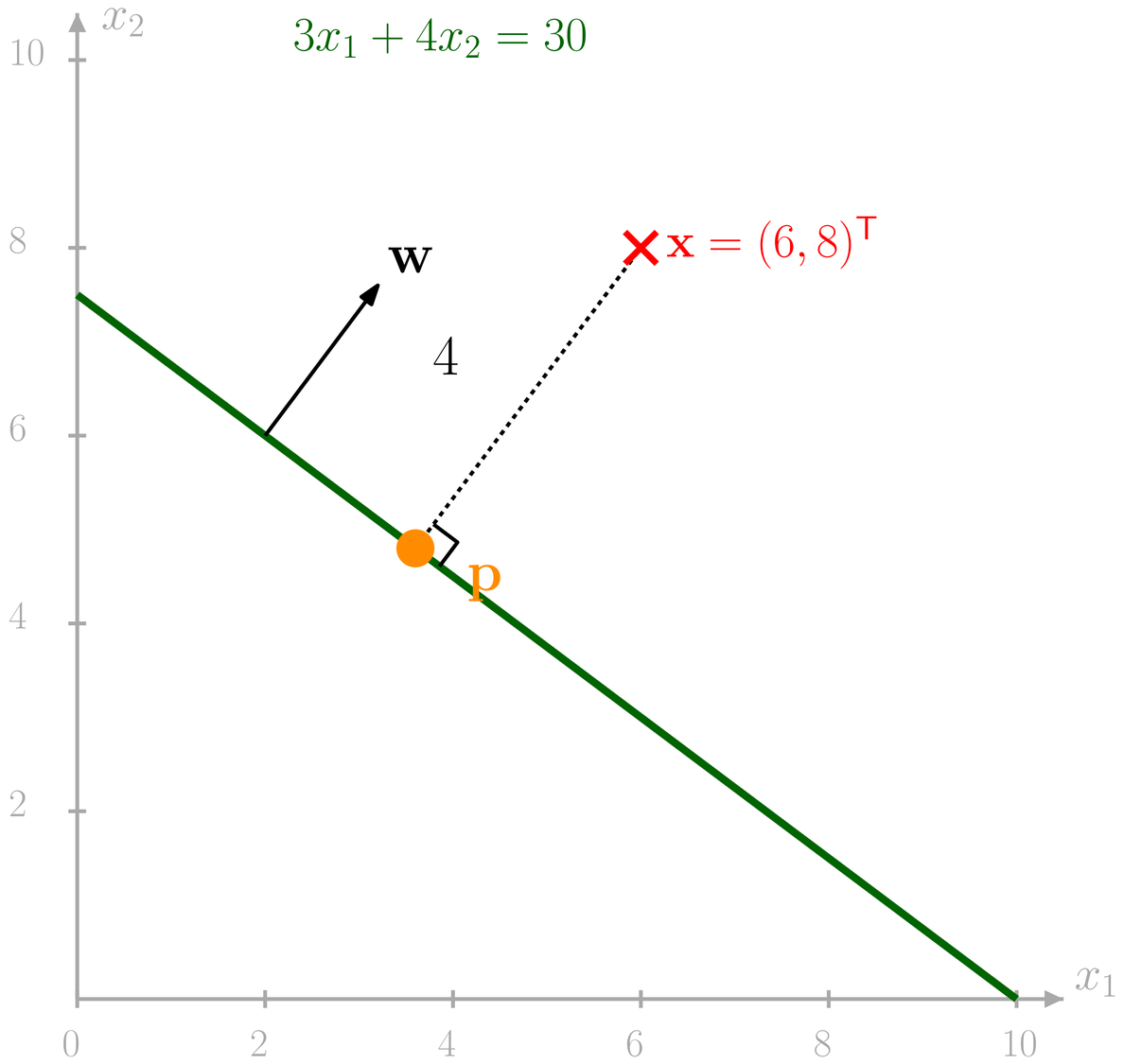
The score is 20 and $\|\mathbf{w}\|_2 = 5$, so the signed distance is 4.

$$\mathbf{p} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} - \frac{20}{25} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 18/5 \\ 24/5 \end{pmatrix}.$$

Rescale to $\mathbf{w}' = (21, 28)^{\mathsf{T}}$, $b' = -210$.

The score becomes 140, but the signed distance remains $140/35 = 4$.

Example 2 (cont'd).



Closest point

$$\mathbf{p} = \begin{pmatrix} 18/5 \\ 24/5 \end{pmatrix}.$$

$\mathbf{x} - \mathbf{p}$ is parallel to $\mathbf{w}$.

$$\|\mathbf{x} - \mathbf{p}\|_2 = 4.$$

The same plot applies to $\mathbf{w}' = (21, 28)^\mathsf{T}$ and $b' = -210$.

Functional and geometric margins. For a labelled datapoint $(\mathbf{x}_i, y_i)$, define

$$m_i \;=\; y_i f_{\mathbf{w},b}(\mathbf{x}_i), \qquad \gamma_i \;=\; \frac{m_i}{\|\mathbf{w}\|_2}.$$

m_i is the functional margin; γ_i is the geometric margin.

If $m_i > 0$, the point is strictly on the correct side.

If $m_i < 0$, it is strictly on the wrong side.

Multiplying by y_i treats the two classes in the same way.

Separability and the dataset margin. The dataset is linearly separable if some $\mathbf{w}, b$ give $m_i > 0$ for every i.

The margin of a separating classifier is

$$\gamma \;=\; \min_{i=1,\ldots,n} \gamma_i.$$

A perturbation $\boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\|_2 < \gamma_i$ cannot change the prediction for that point.

A larger training margin does not by itself guarantee correct predictions on unseen data.

A support vector machine (SVM) maximizes the margin

Hard-margin SVM. Assume both classes occur and the dataset is linearly separable.

Require every functional margin to be at least 1:

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 \\ \text{subject to} \quad & y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \;\geq\; 1 \quad (i = 1, \ldots, n). \end{aligned}$$

These constraints guarantee a geometric margin of at least $1/\|\mathbf{w}\|_2$.

Proposition 2. The hard-margin SVM maximizes the geometric margin. At an optimum,

$$\min_i y_i f_{\mathbf{w},b}(\mathbf{x}_i) \;=\; 1, \qquad \gamma \;=\; \frac{1}{\|\mathbf{w}\|_2}.$$

Support vectors satisfy $y_i f_{\mathbf{w},b}(\mathbf{x}_i) = 1$.

They lie on $f_{\mathbf{w},b}(\mathbf{x}) = 1$ or $f_{\mathbf{w},b}(\mathbf{x}) = -1$.

The two margin boundaries are $2/\|\mathbf{w}\|_2$ apart.

Why minimizing the norm maximizes the margin. For any strict separator, let $t = \min_i y_i f_{\mathbf{w},b}(\mathbf{x}_i) > 0$.

Rescale $\widetilde{\mathbf{w}} = \mathbf{w}/t$ and $\widetilde{b} = b/t$. The smallest functional margin becomes 1, while

$$\gamma \;=\; \frac{t}{\|\mathbf{w}\|_2} \;=\; \frac{1}{\|\widetilde{\mathbf{w}}\|_2}.$$

At an optimum, $t > 1$ is impossible: division by t would keep the constraints satisfied and reduce the squared-norm objective.

Both classes occur, so feasible weights cannot be zero.

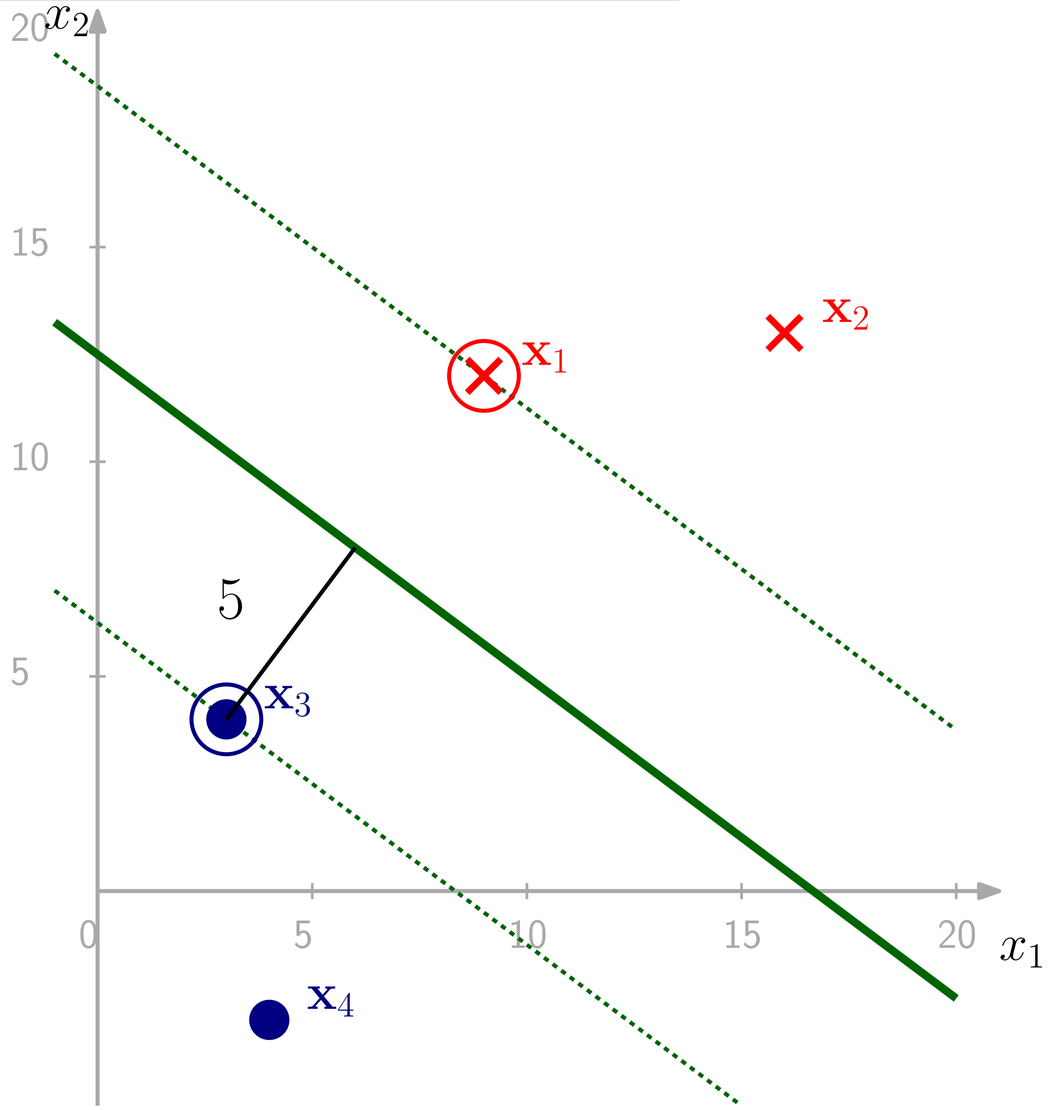
Example 3. Consider four labelled datapoints:

i	$\mathbf{x}_i$	y_i
1	$(9, 12)^\mathsf{T}$	$+1$
2	$(16, 13)^\mathsf{T}$	$+1$
3	$(3, 4)^\mathsf{T}$	-1
4	$(4, -3)^\mathsf{T}$	-1

Take $\mathbf{w} = (3, 4)^\mathsf{T}/25$ and $b = -2$.

Compute the scores, functional margins and geometric margin.

Example 3 (cont'd).



Decision boundary:

$$3x_1 + 4x_2 = 50.$$

Margin boundaries:

$$3x_1 + 4x_2 = 25,$$
$$3x_1 + 4x_2 = 75.$$

Support vectors: $\mathbf{x}_1, \mathbf{x}_3$.

$\|\mathbf{w}\|_2 = 1/5$, so the margin is 5 and the full width is 10.

Example 3 (cont'd). Every feasible pair $\mathbf{w}', b'$ must satisfy the constraints for $\mathbf{x}_1$ and $\mathbf{x}_3$:

$$\langle \mathbf{w}', \mathbf{x}_1 \rangle + b' \ \geq \ 1, \qquad \langle \mathbf{w}', \mathbf{x}_3 \rangle + b' \ \leq \ -1.$$

Subtract, then apply Cauchy–Schwarz:

$$2 \ \leq \ \langle \mathbf{w}', \mathbf{x}_1 - \mathbf{x}_3 \rangle \ \leq \ 10 \|\mathbf{w}'\|_2.$$

Thus $\|\mathbf{w}'\|_2 \geq 1/5$.

Our weights attain this lower bound, so they are optimal.

Soft margins allow violations

Soft-margin SVM. Overlapping classes or imperfect labels can make the hard-margin constraints infeasible.

For $C > 0$, allow penalised violations:

$$\begin{aligned} \text{minimise} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}\xi_i \\ \text{subject to} \quad & y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \;\geq\; 1 - \xi_i, \\ & \xi_i \;\geq\; 0 \qquad (i = 1, \ldots, n). \end{aligned}$$

The slack variables ξ_i permit violations of the unit margin requirement.

Hinge loss. For a functional margin m, define

$$\ell(m) \;=\; \max\{0, 1 - m\}.$$

For fixed $\mathbf{w}, b$, the constraints require

$$\xi_i \;\geq\; 0, \qquad \xi_i \;\geq\; 1 - m_i.$$

Since $C > 0$, the best choice is the smallest permitted value:

$$\xi_i \;=\; \max\{0, 1 - m_i\} \;=\; \ell(m_i).$$

Proposition 3. The soft-margin SVM is equivalently the unconstrained minimisation of

$$\frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n} \ell\big(y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)\big).$$

The first term favours small weights.

The second penalises functional margins below 1.

A correctly classified datapoint can still incur hinge loss.

Interpreting the slack. Use the smallest slack value $\xi_i = \ell(m_i)$ and assume $\mathbf{w} \neq \mathbf{0}$:

slack	location
$\xi_i = 0$	meets the unit margin requirement
$0 < \xi_i < 1$	correct side, inside the margin
$\xi_i = 1$	on the decision boundary
$\xi_i > 1$	strictly on the wrong side

The slack measures a margin violation, not just a classification error.

Example 4. For $\mathbf{w} = (3, 4)^\mathsf{T}/25$ and $b = -2$:

$\mathbf{x}_i$	y_i	$f_{\mathbf{w},b}(\mathbf{x}_i)$	m_i	$\xi_i = \ell(m_i)$
$(12, 16)^\mathsf{T}$	$+1$	2	2	0
$(8, 9)^\mathsf{T}$	$+1$	$2/5$	$2/5$	$3/5$
$(6, 8)^\mathsf{T}$	-1	0	0	1
$(0, 0)^\mathsf{T}$	$+1$	-2	-2	3

The second point is correctly classified but incurs loss $3/5$; the fourth is misclassified and incurs loss 3.

- Larger C: more weight on reducing margin violations.
- Smaller C: more weight on keeping the weights small.
- Changing feature units changes the geometry; consider feature scaling and C together.
- The soft-margin formulation permits $\mathbf{w} = \mathbf{0}$, giving a constant score.

Computing the SVM parameters

- The constrained formulations can be handled by standard quadratic-programming (QP) solvers.
- These SVM problems are *convex*: every local minimum is also a global minimum.
- For the soft-margin SVM, we can instead optimise the hinge-loss objective directly.
- Iterative updates balance smaller weights against corrections for margin violations.

Example 5. At the current $\mathbf{w}, b$, collect the datapoints violating the unit margin:

$$\mathcal{I} \;=\; \{i : y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) < 1\}.$$

For a small step size $0 < \eta < 1$, one possible update is

$$\mathbf{w}_{\text{new}} \;=\; (1-\eta)\mathbf{w} + \eta C \sum_{i \in \mathcal{I}} y_i \mathbf{x}_i,$$

$$b_{\text{new}} \;=\; b + \eta C \sum_{i \in \mathcal{I}} y_i.$$

Both updates use the same $\mathcal{I}$, computed before either parameter changes.

- $(1 - \eta)\mathbf{w}$ shrinks the weights.
- A violating datapoint contributes $\eta C y_i \mathbf{x}_i$ to the weights and $\eta C y_i$ to the bias.
- On its own, this correction increases that point's functional margin by $\eta C(\|\mathbf{x}_i\|_2^2 + 1)$.
- Other points and the weight shrinkage can counteract this increase.

- This is a subgradient method: it handles the corner in hinge loss at margin 1.
- Recompute the margins and repeat, with suitable step sizes and stopping criteria.
- Individual steps need not decrease the objective.
- An arbitrary fixed step size need not converge; detailed convergence conditions are beyond this course.

Takeaways

- An affine score $\langle \mathbf{w}, \mathbf{x} \rangle + b$ gives a binary prediction.
- Divide the score by $\|\mathbf{w}\|_2$ to obtain signed distance.
- The hard-margin SVM seeks the widest separating margin.
- Slack variables and hinge loss allow margin violations.
- Numerical solvers and subgradient methods compute the parameters.